



# Quadratic equations: what we can already solve

## WORKSHEET

Name \_\_\_\_\_

Class \_\_\_\_\_

Date \_\_\_\_\_

### REMEMBER

## From grade 8

Level 1 • threshold

**1.**Solve  $2x^2 = 50$ . Give both roots, separated by a semicolon.

Answer: \_\_\_\_\_

Level 1 • threshold

**2.**Solve  $x^2 - 6x = 0$ .

Answer: \_\_\_\_\_

Level 1 • threshold

**3.**Which of these numbers is a root of  $x^2 - x - 12 = 0$ ?☐ A 3☐ B 12☐ C -4☐ D -3

### WORKED EXAMPLES

## Worked examples

### YOUR TURN

Level 1 • threshold

Write  $4 - x^2 = 3x$  in the form  $ax^2 + bx + c = 0$  with  $a > 0$ . Enter  $a$ ,  $b$  and  $c$ , in this order, separated by semicolons.

Answer: \_\_\_\_\_

### Example 2 • A product equal to zero: $(x - 4)(2x + 3) = 0$

ZERO-PRODUCT RULE

$$x - 4 = 0 \text{ or } 2x + 3 = 0.$$

SOLVE EACH

$$x = 4 \text{ or } 2x = -3, \text{ so } x = -\frac{3}{2}.$$

CHECK • YOUR TURN

#### YOUR TURN

Level 1 • threshold

Solve  $(3x + 2)(x - 5) = 0$ .

Answer:

### Example 3 • A square equal to a number

$$(x + 2)^2 = 16$$

$$16 > 0: x + 2 = 4 \text{ or } x + 2 = -4, \text{ so } x = 2 \text{ or } x = -6.$$

$$(x + 4)^2 = 0 \text{ • YOUR TURN}$$

$$(x + 3)^2 = -4 \text{ • YOUR TURN}$$

#### YOUR TURN

Level 1 • threshold

Solve  $(x - 3)^2 = 25$ .

Answer:

#### CHECK

### Check yourself

Level 1 • threshold

1.

Which of these are quadratic equations? Choose all that apply.

☐  $\frac{4}{x^2} - 1 = 0$

☐  $3^2x + x - 4 = 0$

☐  $x^2 - 7x = 0$

☐  $5 - 2x^2 = 0$

Level 2 • satisfactory

2.

Which of these is an **incomplete** quadratic equation?

☐  $x^2 + x = 1$

☐  $(x - 1)^2 = 0$

☐  $3x^2 = 5x$

☐  $2x^2 + 3x - 5 = 0$

3.

Write  $7 - 3x^2 = 2x$  in the form  $ax^2 + bx + c = 0$  with  $a > 0$ . Enter  $a$ ,  $b$  and  $c$ , in this order.

Answer:

4.

Solve  $(x + 6)(3x - 1) = 0$ .

Answer:

5.

Solve  $(x + 4)^2 = 36$ .

Answer:

6.

Solve  $(x - 5)^2 = -9$ .

- ☐ A  $x = -4$ 
☐ B no real roots
 ☐ C  $x = 5$ 
☐ D  $x = 8$  or  $x = 2$

7.

Ona solves  $x(x - 3) = 4$  like this: " $x = 4$  or  $x - 3 = 4$ , so  $x = 4$  or  $x = 7$ ." What is wrong?

- ☐ A The zero-product rule needs 0 on one side. Here it is 4, so the rule does not apply, and  $x = 7$  is not a root.
 ☐ B She should divide both sides by  $x$ .
 ☐ C Nothing: 4 and 7 are both roots.
 ☐ D She should have written  $x = 0$  or  $x - 3 = 0$ .

## EXAM TASK

### A hidden square

Level 3 · basic

PUPP style · 3 points

The equation  $x^2 - 2x + 1 = k$  is given, where  $k$  is a number.

- a) Write the left side as the square of a binomial.
- b) Solve the equation when  $k = 49$ .
- c) For which values of  $k$  does the equation have no real roots? Explain.

